

## COMMONWEALTH OF AUSTRALIA

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# Acknowledgements

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FIT5226

Multi-agent System &  
Collective Behaviour

Wk 6: Classical Game Theory (Intro)

# Week-to-week structure (tentative)

|                       |    |                                                                     |                          |
|-----------------------|----|---------------------------------------------------------------------|--------------------------|
|                       | 1  | Introduction & Scoping                                              |                          |
|                       | 2  | Macroscopic/mesoscopic system dynamics (deterministic & stochastic) |                          |
| Learning<br>↑<br>↓    | 3  | RL Intro and MDPs                                                   | Bellman Equations        |
|                       | 4  | Policy and Value Iteration                                          | SARSA                    |
|                       | 5  | Q-Learning                                                          | Deep Q-Learning          |
|                       | 6  | Game Theory Intro                                                   |                          |
|                       | 7  | Solution Concepts; Nash                                             | MiniMax-Q, Markov Games  |
| Multi-agent<br>↑<br>↓ | 8  | Population Games                                                    | Evolutionary Game Theory |
|                       | 9  | Social Dilemmas                                                     |                          |
|                       | 10 | Coalition Game Theory                                               |                          |
|                       | 11 | Auctions                                                            |                          |
|                       | 12 | Voting                                                              |                          |

# Classical (Economic) Game theory

John von Neumann & Oskar Morgenstern (1944) *Theory of Games and Economic Behavior*



John von Neumann



Oskar Morgenstern



John Nash

# Game Theory

## History

1921-1927: First approaches (E. Borel)

1928: J. von Neumann: “Zur Theorie der Gesellschaftsspiele”  
in Mathematische Annalen Vol. 100

1944: J. von Neumann and O. Morgenstern:  
“Theory of Games and Economical Behaviour”,  
Princeton University Press  
The defining Volume

1950: PhD John Nash

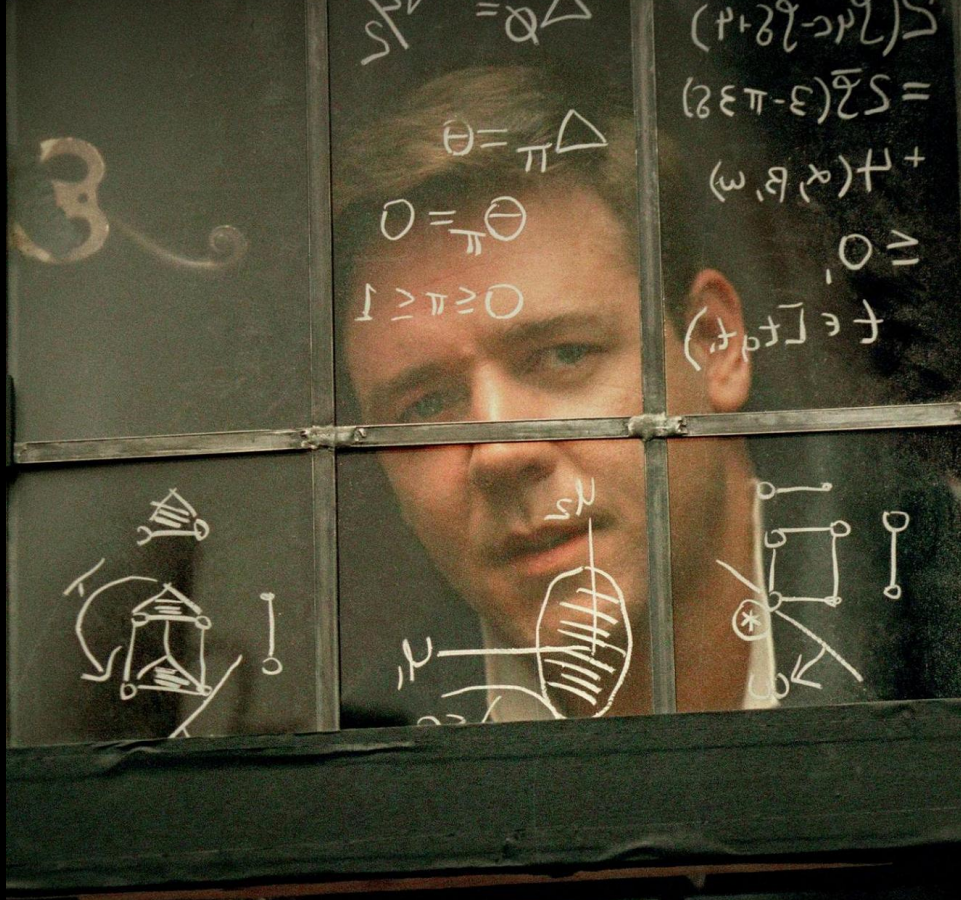
Original Interest: Economy and Warfare --- Strategy selection

# Some Nash Trivia

**Nash's PhD had only 27 pages!**

**Thesis Supervision John Nash (Princeton)**

| Supervisor       | Percentage |
|------------------|------------|
| Albert W. Tucker | 100%       |
| R.J. Duffin      | 0%         |



RUSSELL CROWE  
A BEAUTIFUL MIND

A RON HOWARD FILM



# Non-cooperative Game Theory

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Strategic interaction between **rational**, **self-interested** agents

*non-cooperative* means

- the individual is the basic actor,
- individuals pursue their own interests

cooperative or coalition game theory investigates how agents form coalitions and teams

# Biological and Ecological Games



# Competition for food

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- **It's good to be large**
  - **When beetles of the same size compete, they get equal shares of the food**
  - **When a large beetle competes with a small beetle, the large beetle gets the majority of the food.**
- **But being large is not only positive: large beetles experience less of a fitness benefit from a given quantity of food, since some of it is diverted into maintaining their expensive metabolism**

# Prisoners' Dilemma

An Example of a cooperative, non-zero sum game

Situation: The police arrests two suspects, but has only little evidence. Both suspects are imprisoned in isolation and offered a “deal”. If either acts as witness against the other one he is offered a reduced sentence.

- If both remain silent (cooperate with each other), they gain a little.
- If both talk, both are punished harder as if both remain silent.
- If A “talks” (=defects) and B remains silent (=cooperates with A), A gains most at the expense of B.

|                                               | Prisoner B stays silent ( <i>cooperates</i> ) | Prisoner B betrays ( <i>defects</i> )        |
|-----------------------------------------------|-----------------------------------------------|----------------------------------------------|
| Prisoner A stays silent ( <i>cooperates</i> ) | Each serves 1 year                            | Prisoner A: 3 years<br>Prisoner B: goes free |
| Prisoner A betrays ( <i>defects</i> )         | Prisoner A: goes free<br>Prisoner B: 3 years  | Each serves 2 years                          |

Dilemma: If either player reacts “fully rationally” he/she would always defect (talk). However, the “irrational” hope is that the other one will also cooperate.

Such games become more even more interesting if they are iterated, so that A/B can gain information about B/A’s strategy. (Without it being known when the game ends!)

# A Biological Prisoner's Dilemma



# Self-interested Agents

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## Each Agent

- has their own description of states that they would like to reach
- and chooses their actions according to these target states
- this does *not* necessarily mean that the agent is selfish or acts against others
- agents **independently maximise their own ‘utility’** (“rationality”)



## Example: friends and enemies

Alice chooses her destination for the evening

She can go to the club, the movies, or stay home and watch netflix

her **utility** for these *by herself* is:

club: 100, movies: 50; home: 50

Alice dislikes Bob but likes Carol.

If she goes out, she might bump into either

Bob is at the club 60% of the time, and at the movies otherwise

Carol is at the movies 75% of the time, and at the club otherwise

If Alice runs into Bob at the movies or the club, her utility is only 10

If Alice sees Carol, her utility increases by a factor 1.5

|                |         |         |         |         |
|----------------|---------|---------|---------|---------|
|                | 60%     | 40%     |         |         |
|                | $B = c$ | $B = m$ | $B = c$ | $B = m$ |
| $C = c$<br>25% | 15      | 150     | 50      | 10      |
| $C = m$<br>75% | 10      | 100     | 75      | 15      |
|                | $A = c$ |         | $A = m$ |         |

Alice's **expected** utility for home (h): 50

Alice's expected utility for club (c):

$$0.25(0.6 \cdot 15 + 0.4 \cdot 150) + 0.75(0.6 \cdot 10 + 0.4 \cdot 100) = 51.75.$$

Alice's expected utility for movies (m):

$$0.25(0.6 \cdot 50 + 0.4 \cdot 10) + 0.75(0.6(75) + 0.4(15)) = 46.75.$$

A prefers the club (though Bob is likely to be there and Carol is not), she prefers staying home to the movies (though Carol almost always is there).



# Utility Theory

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Is it sensible (sufficient) to describe an agent's preferences as a utility (numerical) function?

Start from the comparison of only two outcomes by an agent

$O1 \sim O2$ : no preference

$O1 \succeq O2$ : weak preference “ $O1$  is at least as desirable as  $O2$ ”

$O1 \succ O2$ : strict preference

# Lottery

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Define a lottery that assigns probabilities to random outcomes

$$L = [p_1 : o_1, p_2, o_2, \dots, p_n : o_n]$$

We can now consider a whole lottery as an outcome

# Axioms 1/2

1. **Completeness:** a preference is defined between any two outcomes

2. **Transitivity:**  $o_1 \succ o_2 \wedge o_2 \succ o_3 \implies o_1 \succ o_3$

3. **Monotonicity:**

$$o_1 \succ o_2 \wedge p \succ q \implies [p : o_1, 1 - p : o_2] \succ [q : o_1, 1 - q : o_2]$$

4. **Decomposability**

Let  $p(L_i, o_j)$  denote the probability assigned to  $o_j$  in lottery  $L_i$

$$\forall j : p(L_1, o_j) = p(L_2, o_j) . \implies L_1 \sim L_2$$

# Axioms 2/2

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## 5. Substitutability:

$$o_1 \sim o_2 \implies$$

$$[p : o_1, p_3 : o_3, \dots, p_k : o_k] \sim [p : o_2, p_3 : o_3, \dots, p_k : o_k]$$

## 6. Continuity:

$$o_1 \succ o_2 \wedge o_2 \succ o_3 \implies \exists p : o_2 \sim [p : o_1, 1 - p : o_3]$$

# Utility Theorem

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von Neuman & Morgenstern (1944):

if a preference relation satisfies the above axioms,  
there exists a function  $u : O \rightarrow [0,1]$  such that

- $u(o_1) \geq u(o_2) \Leftrightarrow o_1 \succ o_2$
- the agent prefers outcomes that  
maximise the expected value of  $u$

# Example Games: pure competition

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- Players have exactly opposing interests
- Only two players (otherwise, what does exactly opposing mean?)
- Let  $u_i$  be the utility for Player  $i$  for an action profile  $a$

$$\exists c \forall a \in A : u_1(a) + u_2(a) = c$$

# Two-person Zero-sum games

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Two-person Zero-sum games are the best understood form of games  
These games are a special case of pure competition

Two players take turns in making a choice from a set of possibilities

Three possible outcomes:

- draw (no one wins anything)
- player A wins \$ $x$  which are paid by player B
- player B wins \$ $y$  which are paid by player A



# Matching Pennies

Each player shows one side of a coin.

To win, one player must match, the other mismatch

|   | H  | T  |
|---|----|----|
| H | 1  | -1 |
| T | -1 | 1  |

# Rock-Paper-Scissors

|          | Rock | Paper | Scissors |
|----------|------|-------|----------|
| Rock     | 0    | -1    | 1        |
| Paper    | 1    | 0     | -1       |
| Scissors | -1   | 1     | 0        |

# Rock-Paper-Scissors

## ABOUT THE EVENT



## ***2023 RPS TOURNAMENT OF CHAMPIONS***

***for Charity (Cancer Council)***

# Morra



Each Player hides one or two coins.

- each player guesses how many coins the opponent has hidden.
- There is a win if and only if one player guesses correctly, the other one incorrectly.
- In case of a win an amount equal to the total number of hidden coins are paid to the winner.

# Morra Payoff Matrix

Strategies for both players:

|       |                                   | (1/1) | (1/2) | (2/1) | (2/2) |
|-------|-----------------------------------|-------|-------|-------|-------|
| (1/1) | Hide one coin, guess “one coin”   | 0     | 2     | −3    | 0     |
| (1/2) | Hide one coin, guess “two coins”  | −2    | 0     | 0     | 3     |
| (2/1) | Hide two coins, guess “one coin”  | 3     | 0     | 0     | −4    |
| (2/2) | Hide two coins, guess “two coins” | 0     | −3    | 4     | 0     |

As this is zero-sum, a single matrix is enough to denote the outcomes (instead of a bi-matrix notation)

We speak of the “column player” and the “row player”.

- the pay-off is given for the row player (by convention)
- the row player tries to maximize the outcome
- column player tries to minimize the outcome

# Defining Games

Finite,  $n$ -person game:  $\langle N, A, u \rangle$ :

- $N$  is a finite set of  $n$  **players**, indexed by  $i$
- $A = A_1 \times \dots \times A_n$ , where  $A_i$  is the **action set** for player  $i$ 
  - $a \in A$  is an **action profile**, and so  $A$  is the space of action profiles
- $u = \langle u_1, \dots, u_n \rangle$ , a **utility function** for each player, where  $u_i : A \mapsto \mathbb{R}$

Writing a 2-player game as a **matrix**:

- row player is player 1, column player is player 2
- rows are actions  $a \in A_1$ , columns are  $a' \in A_2$
- cells are outcomes, written as a tuple of utility values for each player

# GT - RL Terminology

| <b>Game Theory</b> | <b>Reinforcement Learning</b> |
|--------------------|-------------------------------|
| Player             | Agent                         |
| Action             | Action                        |
| Payoff             | Reward                        |
| Strategy           | Policy                        |
| Action profile     | Joint action                  |

# GT utility/payoff vs RL Reward

- much like the reward in RL, payoff in GT
  - is a scalar value,
  - indicates how well agent is performing
  - shares the underlying assumptions of utility theory
- However, differences are that RL reward
  - is typically cumulative over time
  - depends on action and environment state
    - in RL agents' actions influence the environment state
    - GT (in this form) is stateless
  - can be delayed
    - eg. investing before cashing in later (since we use return)
    - in GT the reward is instantaneous



# More Example Games



Should you send your packets using correctly-implemented **TCP** (which has a “backoff” mechanism) or use a defective implementation (which doesn’t)?

Consider this situation as a two-player game:

- both use a correct implementation: both get 1 ms delay
- one correct, one defective: 4 ms delay for correct, 0 ms for defective
- both defective: both get a 3 ms delay.

# TCP Backoff Games

|   | C      | D      |
|---|--------|--------|
| C | -1, -1 | -4, 0  |
| D | 0, -4  | -3, -3 |

Consider this situation as a two-player game:

- both use a correct implementation: both get 1 ms delay
- one correct, one defective: 4 ms delay for correct, 0 ms for defective
- both defective: both get a 3 ms delay.

Note: the bi-matrix notation gives the direct payoff, so both players try to maximise

# General From: Prisoners' dilemma

|   | C      | D      |
|---|--------|--------|
| C | $a, a$ | $b, c$ |
| D | $c, b$ | $d, d$ |

- any such game with  $c > a > d > b$
- $c$  is called the “Temptation”
- $b$  is called the “Sucker’s reward”

# Traffic Game

|   | L          | R          |
|---|------------|------------|
| L | <i>1,1</i> | <i>0,0</i> |
| R | <i>0,0</i> | <i>1,1</i> |

- which side of the road do you drive on?

# Coordination Games

|   | C      | D      |
|---|--------|--------|
| C | $a, a$ | $b, c$ |
| D | $c, b$ | $d, d$ |

- players have aligned interests,
- no conflict
- $\forall a \in A, \forall i, j, u_i(a) = u_j(a)$
- this is not cooperative game theory (which is about team formation)

# Battle of the Sexes

|   | B           | F           |
|---|-------------|-------------|
| B | <i>2, 1</i> | <i>0, 0</i> |
| F | <i>0, 0</i> | <i>1, 2</i> |

- combines elements of co-operation and of competition

# Take home lessons

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- Game Theory provides a suitable tool to analyse the interactions of multiple agents
- The basis of game theory is the assumption of rational behaviour
- “Rational” is defined by certain assumptions about reasoning and behaviour of the players, including this of utility theory (these are not based in psychology but logical assumption)
- Classical game theory only analyses outcomes of games, it does not talk about how players “learn” to play a game
- Part of the GT toolbox is a “zoo” of paradigmatic games that encapsulate typical scenarios (for example, pure competition, coordination, mixes of these, conflicts between individual advantage and good group outcomes etc).
- Two-person zero-sum games model the simplest form of direct and pure competition